operating and differential levers on the upper end of the vertical shaft.

The cover assembly serves as a carrier for the throttle control lever, stop lever, control cam, and covers the top of the control housing.

The lower portion of governor is lubricated by means of a slinger attached to front end of blower lower rotor shaft. This slinger dips into a well of oil dammed up in blower housing cover and throws the oil onto all parts within governor weight housing. The upper portion of the governor, including the vertical shaft bearings and the control mechanism, is lubricated partly by splash from oil slinger on weight carrier shaft and partly by return oil through the vertical shaft housing from cylinder head.

OPERATION

The limiting speed mechanical governor provides full fuel for starting when the throttle control lever is in the idle position. Immediately after starting, the governor moves the injector racks out to the position required for idling. Engine speed may then be manually controlled, within the governed limits, by the operator.

In the low speed range the governor weights operate against the low speed spring. The engine idle speed is determined by the engine speed required to rotate the governor weights with sufficient force to form a balance with the low speed spring. Adjustment of the low speed spring adjusting screw will change the compression of the low speed spring and thus raise or lower the engine idle speed.

An increase in engine speed will continue to rotate the weights faster and compress the low speed spring until the spring cap is tight against the spring plunger. This removes the low speed spring from operation and the governor then is in the intermediate speed range where speed control is entirely manual.

When engine speed is increased until rotation of the weights overcomes the pre-load on the high speed spring the governor will retard the injector racks and engine speed will be reduced to the maximum governed speed. Adjustment of the high speed spring retainer will change the compression of the high speed spring and, thus, will raise or lower the maximum governed speed of the engine.

The outward motion of the governor weights, is changed into linear motion by the bell crank extension on each weight and is applied to the riser. The riser transmits this force through the ball thrust bearing

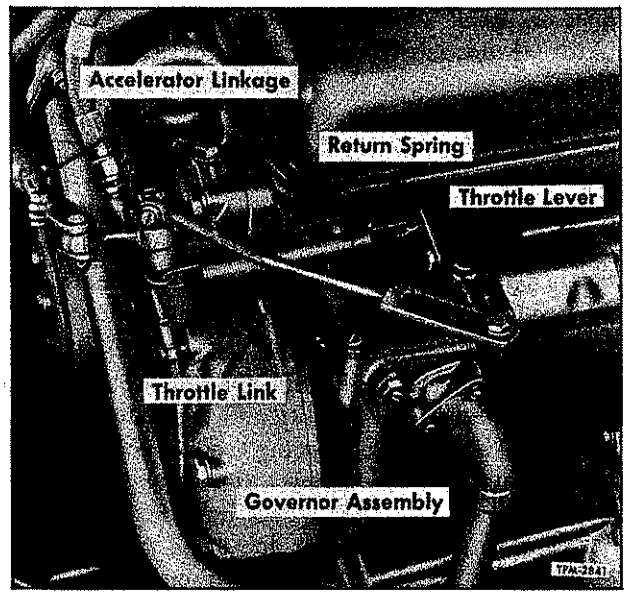


Figure 50—Limiting Speed Governor Mounting—Transit Coach

to a forked lever on the lower end of the vertical shaft. An operating lever is secured to the other end of the vertical shaft. One end of this operating lever bears against the high and low speed springs as described above, while the other end provides a changing fulcrum for the differential lever. The differential lever permits both manual and governor control of the injector racks. Thus, the limiting speed governor will control the idle and maximum engine speed while permitting manual control of engine speeds in the intermediate range.

MAINTENANCE ON VEHICLE

Governor faults are usually manifest in speed variations of the engine, but it does not necessarily follow that all such speed variations indicate governor faults. Therefore, when improper speed variations appear, the following procedure should be carried out.

1. Check the load to be sure that speed changes observed are not the result of load fluctuations.
2. If the load is uniform, carefully check engine to be sure that all cylinders are firing properly.
3. See that no bind exists in any of the governor mechanism or operating linkage between governor and engine; also, that no bind is manifest in injector control rack shaft or its mounting brackets.

If all cylinders are not firing properly, remove and check the injector from the faulty cylinder. The